Sonder - Iteration 1 database tables

41 tables, 244 columns, grouped by domain. PK = primary key. FK -> table.column shows the table and column a foreign key points to. All tables live in the **sonder** schema.

Users and access (3 tables)

app_user

One account per person with a single role; deactivation keeps history.

- id (PK)
- email
- role
- display_name
- locale
- time_zone
- status
- created_at

password_credential

The account's password hash, kept apart from the profile.

- id (PK)
- user_id (FK -> app_user.id)
- password_hash
- changed_at

auth_session

A login session: hashed token, expiry and revocation.

- id (PK)
- user_id (FK -> app_user.id)
- token_hash
- created_at
- expires_at
- revoked_at

Classes (4 tables)

classroom

A class a teacher creates and students join with a unique join code.

- id (PK)
- name
- note
- join_code
- grade_level
- time_zone
- status
- created_at

class_teacher

Which teachers teach a class (many-to-many); departure keeps attribution.

- id (PK)
- class_id (FK -> classroom.id)
- teacher_id (FK -> app_user.id)
- status
- assigned_at

enrolment

A student's membership of one class; one row per (class, student).

- id (PK)
- class_id (FK -> classroom.id)
- student_id (FK -> app_user.id)
- status
- joined_at

class_subject

Which subjects a class studies; a run needs its class to teach the pack's subject.

- id (PK)
- class_id (FK -> classroom.id)
- subject_id (FK -> subject.id)

Math content pack (14 tables)

subject

A subject (Mathematics now; Physics and Chemistry later) that scopes codes and IDs.

- id (PK)
- code
- name

content_version

One versioned content pack per subject; frozen packs are immutable and carry the pack file hash.

- id (PK)
- subject_id (FK -> subject.id)
- version_code
- note
- pack_sha256
- status
- created_at

topic

Stable topic identity within a subject (external ID).

- id (PK)
- subject_id (FK -> subject.id)
- external_id

topic_revision

The topic as it appears in one pack version (name, blurb).

- id (PK)
- topic_id (FK -> topic.id)
- content_version_id (FK -> content_version.id)
- name
- blurb
- grade8_in_scope

misconception

Stable misconception identity: code unique per subject, plus external ID such as MATH-1379.

- id (PK)
- subject_id (FK -> subject.id)
- code
- external_id

misconception_revision

The misconception's definition and engine readiness in one pack version.

- id (PK)
- misconception_id (FK -> misconception.id)
- content_version_id (FK -> content_version.id)
- name
- description
- cognitive_origin
- status
- engine_tier

topic_misconception

Which misconceptions a topic covers: its home topic (with a coverage target) or a cross-topic membership (with a reason).

- id (PK)
- topic_revision_id (FK -> topic_revision.id)
- misconception_revision_id (FK -> misconception_revision.id)
- membership
- target_minimum
- old_topic_label
- review_note
- disposition

source_reference

A citable source with its licence.

- id (PK)
- citation_key
- citation
- url
- licence
- scope_note

misconception_source
Sources behind a misconception definition.

- id (PK)
- misconception_revision_id (FK -> misconception_revision.id)
- source_id (FK -> source_reference.id)
- locator

item
A four-option question: stem, topic, target misconception, author, LLM-assistance disclosure, review state, twin group for A/B twins.

- id (PK)
- topic_revision_id (FK -> topic_revision.id)
- target_misconception_revision_id (FK -> misconception_revision.id)
- external_id
- stem
- batch
- difficulty_est
- origin
- author_id (FK -> app_user.id)
- llm_assisted
- assistance_note
- review_status
- twin_group

item_option
One of the four labelled options (A-D); exactly one is correct.

- id (PK)
- item_id (FK -> item.id)
- label
- external_id
- option_text
- is_correct

option_misconception
Q-matrix: a wrong option's link to up to three misconceptions, with rationale and link grade (S sound, T stretch).

- id (PK)
- option_id (FK -> item_option.id)
- misconception_revision_id (FK -> misconception_revision.id)
- rationale
- link_grade

item_source
Sources behind an item (required at freeze for curated items).

- id (PK)
- item_id (FK -> item.id)
- source_id (FK -> source_reference.id)
- locator

item_review
A recorded review decision (approved or rejected) by someone other than the author.

- id (PK)
- item_id (FK -> item.id)
- review_round
- reviewer_id (FK -> app_user.id)
- decision
- note
- decided_at

Diagnostic units (3 tables)

diagnostic_unit
The session scope (spec v2): a named unit of a pack with its own hypothesis list and item list.

- id (PK)
- content_version_id (FK -> content_version.id)
- unit_code
- name

unit_hypothesis
A unit's misconception hypotheses in slot order 0..k-1 (NONE and UNMODELLED follow implicitly).

- id (PK)
- unit_id (FK -> diagnostic_unit.id)
- hypothesis_id (FK -> hypothesis.id)
- slot

unit_item
Which items a unit may ask; an item may belong to several units, and all its links must be unit hypotheses.

- id (PK)
- unit_id (FK -> diagnostic_unit.id)
- item_id (FK -> item.id)

Engine model (frozen inputs) (6 tables)

hypothesis
The engine's candidate states in a pack: each misconception plus NONE and UNMODELLED.

- id (PK)
- content_version_id (FK -> content_version.id)
- kind
- misconception_revision_id (FK -> misconception_revision.id)

engine_release
An engine code build and its selection policy, pinned by every run.

- id (PK)
- release_code
- code_hash
- selection_policy
- created_at

engine_params
One versioned, hashed engine parameter file: slip, guess, reserved priors, thresholds, support rules, budget, K_MAX, link weights (all assumed).

- id (PK)
- params_version
- params_sha256
- slip
- guess
- prior_none
- prior_unmodelled
- theta
- theta_unmodelled
- min_support
- min_support_none
- delta_sep
- eig_floor
- default_budget
- k_max
- w_sound
- w_stretch
- created_at

response_model
A versioned set of answer likelihoods for a frozen pack, built from one engine_params file.

- id (PK)
- content_version_id (FK -> content_version.id)
- engine_params_id (FK -> engine_params.id)
- model_code
- source
- calibration_note
- status
- created_at

response_parameter
Slip and guess assumptions per hypothesis in a response model.

- id (PK)
- response_model_id (FK -> response_model.id)
- hypothesis_id (FK -> hypothesis.id)
- slip
- guess

response_probability
P(option | hypothesis) for every approved item; four values summing to 1.

- id (PK)
- response_model_id (FK -> response_model.id)
- hypothesis_id (FK -> hypothesis.id)
- option_id (FK -> item_option.id)
- probability

Diagnostic runs (10 tables)

diagnostic_session
One diagnostic run for one enrolment on one diagnostic unit, pinned to a response model and engine release; outcome status.

- id (PK)
- enrolment_id (FK -> enrolment.id)
- response_model_id (FK -> response_model.id)
- engine_release_id (FK -> engine_release.id)
- unit_id (FK -> diagnostic_unit.id)
- question_budget
- request_key
- started_at
- ended_at
- status

session_hypothesis
Each hypothesis in a run (exactly its unit's list plus NONE and UNMODELLED), its slot and its prior probability.

- id (PK)
- session_id (FK -> diagnostic_session.id)
- hypothesis_id (FK -> hypothesis.id)
- slot
- prior_probability

session_question
A question shown at a step, with why it was chosen (entropy, expected information gain).

- id (PK)
- session_id (FK -> diagnostic_session.id)
- step_index
- item_id (FK -> item.id)
- presented_at
- selection_reason
- entropy_before
- expected_ig

question_candidate
The ranked EIG of every candidate item considered for a shown question; rank 1 is the shown item.

- id (PK)
- session_question_id (FK -> session_question.id)
- item_id (FK -> item.id)
- rank
- eig

session_step
The student's answer to a shown question and the engine's answer note; append-only, idempotent by request key.

- id (PK)
- session_id (FK -> diagnostic_session.id)
- step_index
- session_question_id (FK -> session_question.id)
- answer_option_id (FK -> item_option.id)
- answered_at
- request_key
- calculation_ms
- answer_note

step_posterior
The posterior probability of each hypothesis after an answer.

- id (PK)
- step_id (FK -> session_step.id)
- session_hypothesis_id (FK -> session_hypothesis.id)
- probability

diagnosis
The engine's conclusion for a diagnosed run.

- id (PK)
- session_id (FK -> diagnostic_session.id)
- created_at

diagnosis_candidate
Rank 1 and rank 2 hypotheses of a diagnosis (confidence read from the last posterior).

- id (PK)
- diagnosis_id (FK -> diagnosis.id)
- session_hypothesis_id (FK -> session_hypothesis.id)
- rank

escalation
The engine's explicit refusal: reason and leading hypothesis (no separator, budget exhausted, unmodelled).

- id (PK)
- session_id (FK -> diagnostic_session.id)
- reason
- leader_session_hypothesis_id (FK -> session_hypothesis.id)
- created_at

escalation_candidate
The tied set behind a refusal (two or more for no separator), ranked; includes the leader.

- id (PK)
- escalation_id (FK -> escalation.id)
- session_hypothesis_id (FK -> session_hypothesis.id)
- rank

Audit (1 table)

audit_event
Append-only log of important inserts, updates and freezes (who, what, when).

- id (PK)
- actor_id (FK -> app_user.id)
- event_type
- relation_name
- row_id
- request_key
- occurred_at
- detail